



Welcome from the engineering team!

Thank you for your interest in joining our team! We look forward to reviewing your solution to our challenge. When you are ready, email a link to your public GitHub repo to jobs@t19s.com.

- Use any front-end and back-end technology.
- There are no tricks or *gotchas* here.
- No need to work on this for more than a couple hours or less.
- To evaluate the project, we will **git clone** your repository, perform any installation and deployment instructions provided in your repo's README, and then demo a local deployment of your app to our engineering team.
- Feel free to use any tools.

The Community Voices challenge

1. Pick any community with an online presence. This online presence should be active enough for there to be frequent discussions throughout any given week. For example:
 - a. The Argentine tango community and their many local calendars and Facebook discussion groups
 - b. Indie game makers who speak on Discord
 - c. Redditors who create maps of fantasy realms
 - d. Any other community you like with an active online presence, whether focused locally or globally.
2. Create a web or mobile app that generates a **Community Voices Document** that discusses what that community has talked about in the past week and what you predict that community will talk about in the week to come. This can be a Word document, a Power Point presentation, a web page, or anything else that relies on text or any other visualizations as you see fit.

3. Use [RAG](#) to empower the generation of your **Community Voices Document**.
 - a. Consider using either a vector database or a vectorized database table in a relational database (e.g. Postgresql).
 - b. Consider showing a flattened visualization of the embeddings you create.
 - c. Upon retrieving any embedding, consider keeping stats on which embeddings get retrieved the most.
4. Create an automated way to fill up your vector data store with the voices of the community you have chosen.
 - a. Consider using web crawlers or agentic workflows to automatically ingest that community's weekly data
 - b. Think about ways to get around overly large amounts of data
5. A/B test the results: how does an LLM handle the generation of this **Community Voices Document** without your RAG-empowered system? How does your RAG-empowered system generate that document instead? Please provide a comparison within your solution.